

AI ASSISTED CODING

ASSIGNMENT – 11.5

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Task Description #1 – Stack Implementation

Task: Use AI to generate a Stack class with push, pop, peek, and is_empty methods.

Sample Input Code:

```
class Stack: pass
```

Expected Output:

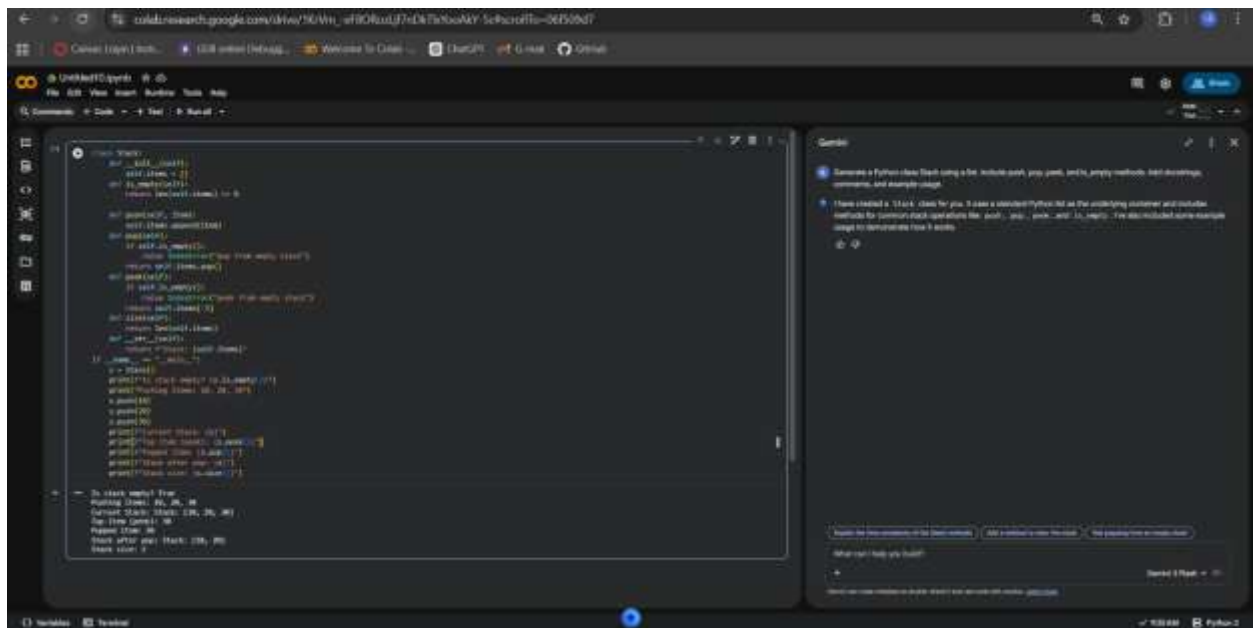
- A functional stack implementation with all required methods and Docstring.

Prompt :

Generate a Python class Stack using a list.

Include push, pop, peek, and is_empty methods.

Add docstrings, comments, and example usage.



Task Description #2 – Queue Implementation

Task: Use AI to implement a Queue using Python lists.

Sample Input Code:

```
class Queue:
```

```
pass
```

Expected Output:

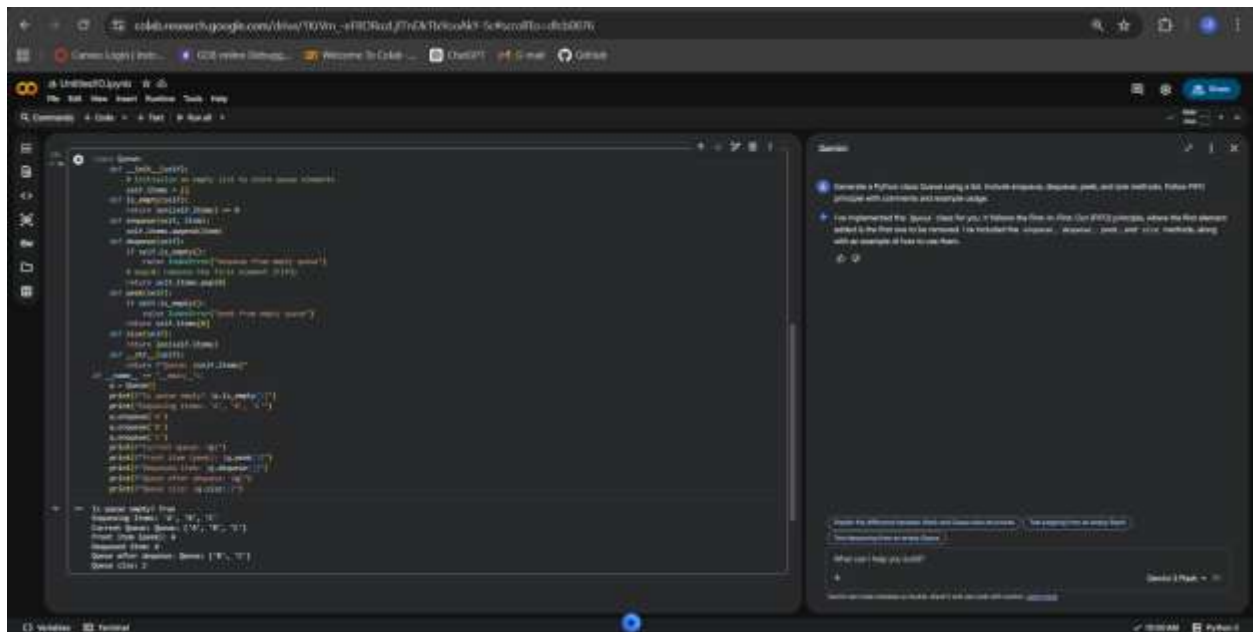
- FIFO-based queue class with enqueue, dequeue, peek, and size methods.

Prompt used :

Generate a Python class Queue using a list.

Include enqueue, dequeue, peek, and size methods.

Follow FIFO principle with comments and example usage.



Task Description #3 – Linked List

Task: Use AI to generate a Singly Linked List with insert and display methods.

Sample Input Code:

```
class Node:
```

```
pass
```

```
class LinkedList:
```

```
pass
```

Expected Output:

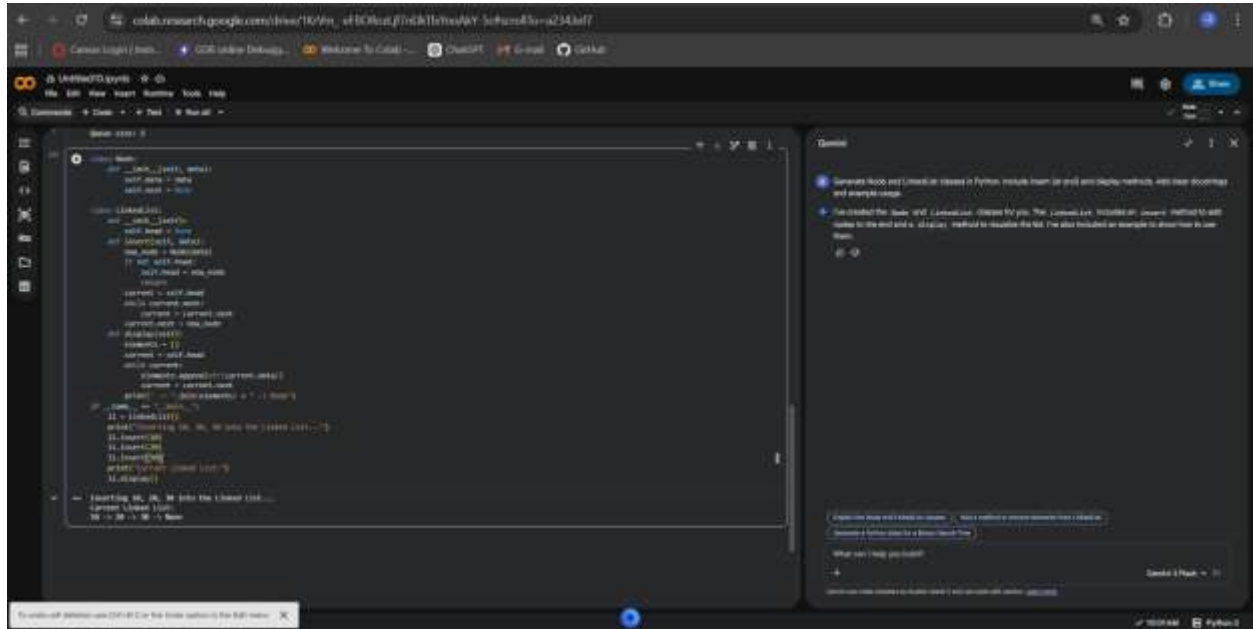
- A working linked list implementation with clear method documentation.

Prompt used :

Generate Node and LinkedList classes in Python.

Include insert (at end) and display methods.

Add clear docstrings and example usage.



Task Description #4 – Hash Table

Task: Use AI to implement a hash table with basic insert, search, and delete methods.

Sample Input Code:

class Hash Table:

pass

Expected Output:

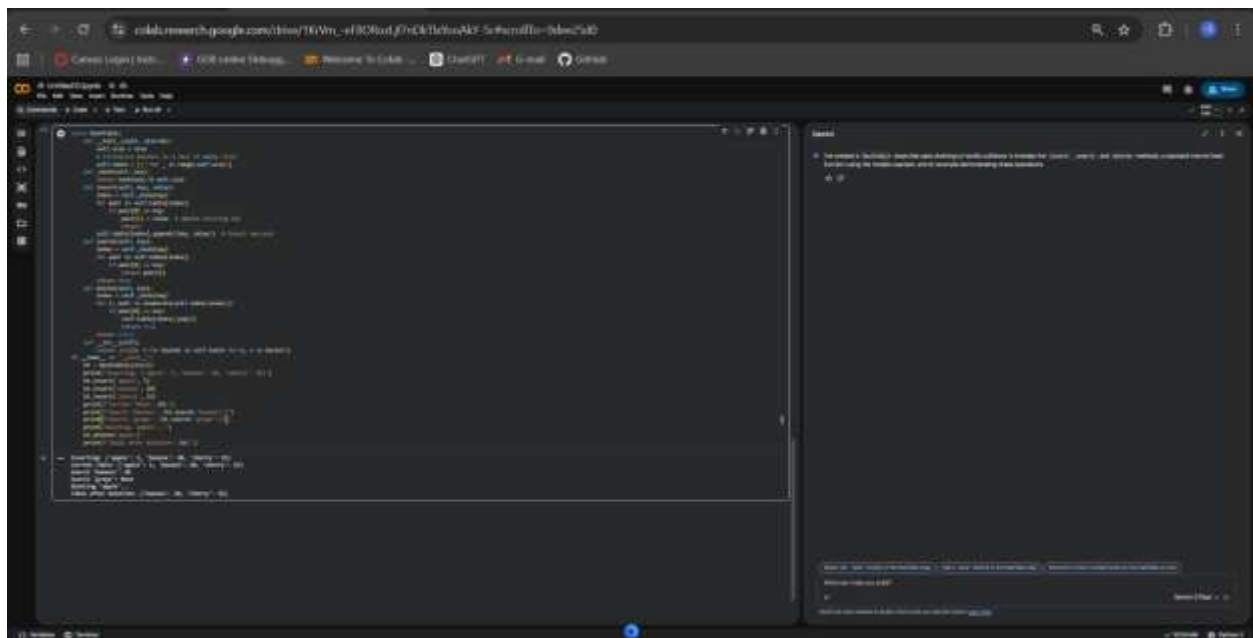
- Collision handling using chaining, with well-commented methods.

Prompt used :

Generate a Python HashTable class using chaining.

Include insert, search, and delete methods.

Add comments, hash function, and example usage.



Task Description #5 – Graph Representation

Task: Use AI to implement a graph using an adjacency list.

Sample Input Code:

```
class Graph : pass
```

Expected Output:

- Graph with methods to add vertices, add edges, and display connections.

Prompt Used :

Generate a Python Graph class using adjacency list.

Include add_vertex, add_edge, and display methods.

Add comments and example usage.

the list below:

- o Stack
- o Queue
- o Priority Queue
- o Linked List
- o Binary Search Tree (BST)
- o Graph
- o Hash Table
- o Deque

- Justify your choice in 2–3 sentences per feature.
- Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

- A table mapping feature → chosen data structure → justification.
- A functional Python program implementing the chosen feature with comments and docstrings.

Prompt Used : (FOR TABLE)

Generate a formatted table for a Smart Hospital Management System.

Features:

1. Patient Check-In System
2. Emergency Case Handling
3. Medical Records Storage
4. Doctor Appointment Scheduling

5. Hospital Room Navigation

Choose the most appropriate data structure from:

Stack, Queue, Priority Queue, Linked List, Binary Search Tree (BST),
Graph, Hash Table, Deque.

Requirements:

- Create a table: Feature | Data Structure | Justification
- Provide 2–3 sentence justification for each feature.
- Keep explanations clear and concise.

Prompt Used : (FOR CODE)

Generate a working Python program implementing Emergency Case Handling using a Priority Queue.

Requirements:

- Use heapq module.
- Each patient has name and severity (lower number = higher priority).

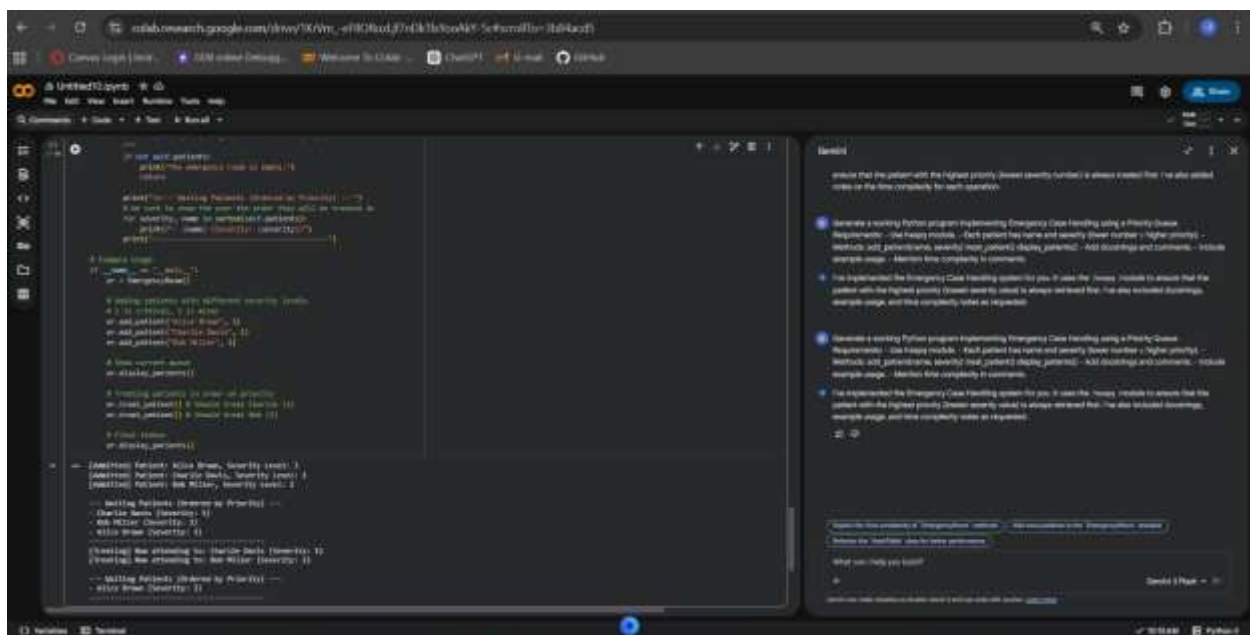
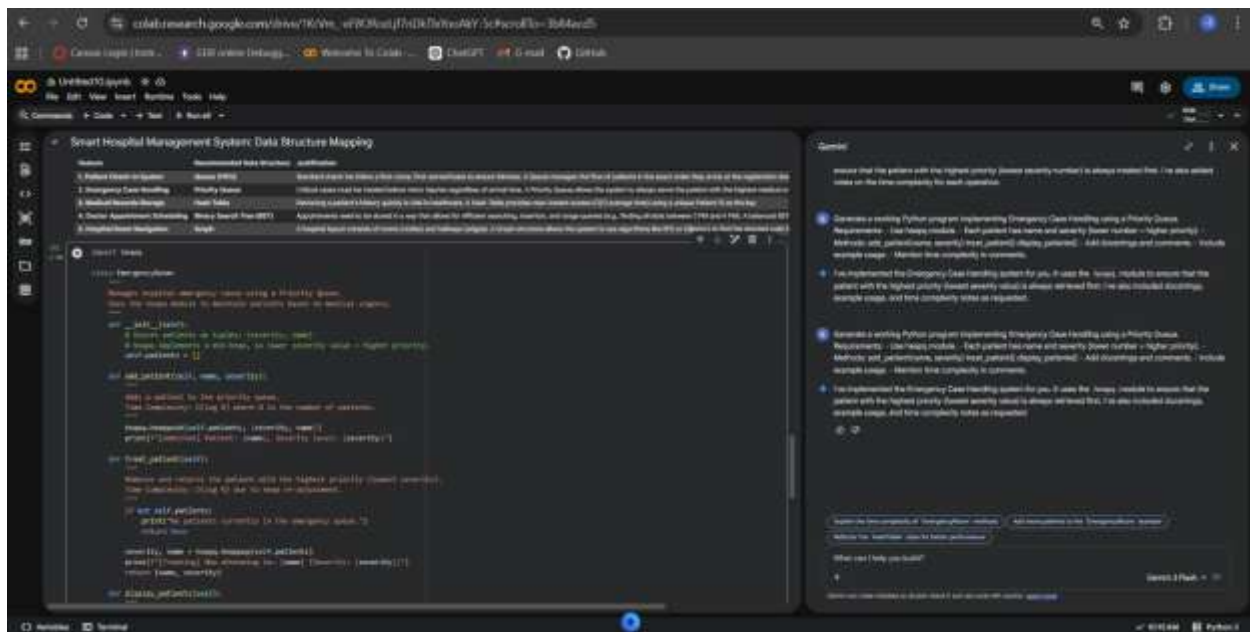
- Methods:

 add_patient(name, severity)

 treat_patient()

 display_patients()

- Add docstrings and comments.
- Include example usage.
- Mention time complexity in comments.



Task Description #7: Smart City Traffic Control System

A city plans a Smart Traffic Management System that includes:

1. Traffic Signal Queue – Vehicles waiting at signals.
2. Emergency Vehicle Priority Handling – Ambulances and fire trucks prioritized.

3. Vehicle Registration Lookup – Instant access to vehicle details.

4. Road Network Mapping – Roads and intersections connected logically.

5. Parking Slot Availability – Track available and occupied slots.

Student Task

- For each feature, select the most appropriate data structure from the list below:

- o Stack

- o Queue

- o Priority Queue

- o Linked List

- o Binary Search Tree (BST)

- o Graph

- o Hash Table

- o Deque

- Justify your choice in 2–3 sentences per feature.

- Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

- A table mapping feature → chosen data structure → justification.

- A functional Python program implementing the chosen feature with comments and docstrings.

Prompt Used : (FOR TABLE)

Generate a formatted table for a Smart Traffic Management System.

Features:

1. Traffic Signal Queue
2. Emergency Vehicle Priority Handling
3. Vehicle Registration Lookup
4. Road Network Mapping
5. Parking Slot Availability

Choose the most appropriate data structure from:

Stack, Queue, Priority Queue, Linked List, Binary Search Tree (BST),
Graph, Hash Table, Deque.

Requirements:

- Format as a table: Feature | Data Structure | Justification
- Provide 2–3 sentence justification per feature.
- Keep explanations simple and clear.

PROMPT USED : (FOR CODE)

Generate a working Python program implementing Vehicle Registration
Lookup

using a Hash Table.

Requirements:

- Store vehicle_number as key.
- Store owner_name and vehicle_type as values.
- Methods:

register_vehicle()

```
delete_vehicle()
```

-
- The screenshot shows a web browser window displaying the GitHub repository page for 'UnleashIO/unleash'. The browser's address bar shows the URL 'https://github.com/UnleashIO/unleash'. The page header includes the repository name and a search bar. The main content area displays the repository's README file, which is written in Markdown. The README describes the project as a 'Feature Toggle' and provides information about its usage, including a link to the 'Unleash' website. The page also shows a list of contributors and a section for 'Getting Started'.

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- The screenshot shows a web browser displaying a GitHub repository for a project titled "Smart Traffic Management System: Data Structure Mapping". The repository is owned by "vishal1604" and contains a single file named "main.py". The "main.py" file is a Python script that defines a data structure for a traffic management system. It includes a list of vehicles, a dictionary of traffic flow, and a list of accidents. The README file provides a detailed description of the system's data structure mapping, including vehicle registration, traffic flow, and accident reporting. The README also includes a table summarizing the data structure mapping.
- Smart Traffic Management System: Data Structure Mapping**
- This repository contains a Python script that defines a data structure for a traffic management system. The data structure is a dictionary where the keys are the names of the vehicles and the values are lists of dictionaries representing the vehicle's registration details, traffic flow, and accident history.
- The data structure is defined as follows:
- ```
data = {
 "Vehicle 1": [
 {"Registration": "MH-123", "Owner": "John Doe", "Type": "Car"},
 {"Flow": "North", "Volume": 100, "Direction": "North"},
 {"Accident": "Accident 1", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 2": [
 {"Registration": "MH-456", "Owner": "Jane Smith", "Type": "Truck"},
 {"Flow": "South", "Volume": 50, "Direction": "South"},
 {"Accident": "Accident 2", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 3": [
 {"Registration": "MH-789", "Owner": "Mike Johnson", "Type": "Bike"},
 {"Flow": "East", "Volume": 20, "Direction": "East"},
 {"Accident": "Accident 3", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 4": [
 {"Registration": "MH-012", "Owner": "Sarah Brown", "Type": "Car"},
 {"Flow": "West", "Volume": 150, "Direction": "West"},
 {"Accident": "Accident 4", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 5": [
 {"Registration": "MH-345", "Owner": "David Wilson", "Type": "Truck"},
 {"Flow": "North", "Volume": 75, "Direction": "North"},
 {"Accident": "Accident 5", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 6": [
 {"Registration": "MH-678", "Owner": "Emily Davis", "Type": "Bike"},
 {"Flow": "South", "Volume": 30, "Direction": "South"},
 {"Accident": "Accident 6", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 7": [
 {"Registration": "MH-901", "Owner": "Chris Miller", "Type": "Car"},
 {"Flow": "East", "Volume": 120, "Direction": "East"},
 {"Accident": "Accident 7", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 8": [
 {"Registration": "MH-234", "Owner": "Lisa Anderson", "Type": "Truck"},
 {"Flow": "West", "Volume": 60, "Direction": "West"},
 {"Accident": "Accident 8", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 9": [
 {"Registration": "MH-567", "Owner": "Tom Taylor", "Type": "Bike"},
 {"Flow": "North", "Volume": 40, "Direction": "North"},
 {"Accident": "Accident 9", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 10": [
 {"Registration": "MH-890", "Owner": "Nina White", "Type": "Car"},
 {"Flow": "South", "Volume": 90, "Direction": "South"},
 {"Accident": "Accident 10", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 11": [
 {"Registration": "MH-101", "Owner": "Daniel Black", "Type": "Truck"},
 {"Flow": "East", "Volume": 110, "Direction": "East"},
 {"Accident": "Accident 11", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 12": [
 {"Registration": "MH-432", "Owner": "Olivia Green", "Type": "Bike"},
 {"Flow": "West", "Volume": 25, "Direction": "West"},
 {"Accident": "Accident 12", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 13": [
 {"Registration": "MH-765", "Owner": "Liam Brown", "Type": "Car"},
 {"Flow": "North", "Volume": 80, "Direction": "North"},
 {"Accident": "Accident 13", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 14": [
 {"Registration": "MH-098", "Owner": "Sophia White", "Type": "Truck"},
 {"Flow": "South", "Volume": 55, "Direction": "South"},
 {"Accident": "Accident 14", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 15": [
 {"Registration": "MH-321", "Owner": "Noah Black", "Type": "Bike"},
 {"Flow": "East", "Volume": 35, "Direction": "East"},
 {"Accident": "Accident 15", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 16": [
 {"Registration": "MH-654", "Owner": "Ava Green", "Type": "Car"},
 {"Flow": "West", "Volume": 130, "Direction": "West"},
 {"Accident": "Accident 16", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 17": [
 {"Registration": "MH-987", "Owner": "Ethan Brown", "Type": "Truck"},
 {"Flow": "North", "Volume": 65, "Direction": "North"},
 {"Accident": "Accident 17", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 18": [
 {"Registration": "MH-210", "Owner": "Mia White", "Type": "Bike"},
 {"Flow": "South", "Volume": 45, "Direction": "South"},
 {"Accident": "Accident 18", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 19": [
 {"Registration": "MH-543", "Owner": "Lucas Black", "Type": "Car"},
 {"Flow": "East", "Volume": 105, "Direction": "East"},
 {"Accident": "Accident 19", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 20": [
 {"Registration": "MH-876", "Owner": "Charlotte Green", "Type": "Truck"},
 {"Flow": "West", "Volume": 70, "Direction": "West"},
 {"Accident": "Accident 20", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 21": [
 {"Registration": "MH-109", "Owner": "Benjamin Brown", "Type": "Bike"},
 {"Flow": "North", "Volume": 50, "Direction": "North"},
 {"Accident": "Accident 21", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 22": [
 {"Registration": "MH-432", "Owner": "Emily White", "Type": "Car"},
 {"Flow": "South", "Volume": 85, "Direction": "South"},
 {"Accident": "Accident 22", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 23": [
 {"Registration": "MH-765", "Owner": "Alexander Black", "Type": "Truck"},
 {"Flow": "East", "Volume": 115, "Direction": "East"},
 {"Accident": "Accident 23", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 24": [
 {"Registration": "MH-098", "Owner": "Sofia Green", "Type": "Bike"},
 {"Flow": "West", "Volume": 30, "Direction": "West"},
 {"Accident": "Accident 24", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 25": [
 {"Registration": "MH-321", "Owner": "Mason Brown", "Type": "Car"},
 {"Flow": "North", "Volume": 95, "Direction": "North"},
 {"Accident": "Accident 25", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 26": [
 {"Registration": "MH-654", "Owner": "Isabella White", "Type": "Truck"},
 {"Flow": "South", "Volume": 60, "Direction": "South"},
 {"Accident": "Accident 26", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 27": [
 {"Registration": "MH-987", "Owner": "Jacob Black", "Type": "Bike"},
 {"Flow": "East", "Volume": 40, "Direction": "East"},
 {"Accident": "Accident 27", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 28": [
 {"Registration": "MH-210", "Owner": "Mila Green", "Type": "Car"},
 {"Flow": "West", "Volume": 125, "Direction": "West"},
 {"Accident": "Accident 28", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 29": [
 {"Registration": "MH-543", "Owner": "Elijah Brown", "Type": "Truck"},
 {"Flow": "North", "Volume": 75, "Direction": "North"},
 {"Accident": "Accident 29", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 30": [
 {"Registration": "MH-876", "Owner": "Aria White", "Type": "Bike"},
 {"Flow": "South", "Volume": 55, "Direction": "South"},
 {"Accident": "Accident 30", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 31": [
 {"Registration": "MH-109", "Owner": "Carter Black", "Type": "Car"},
 {"Flow": "East", "Volume": 100, "Direction": "East"},
 {"Accident": "Accident 31", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 32": [
 {"Registration": "MH-432", "Owner": "Evelyn Green", "Type": "Truck"},
 {"Flow": "West", "Volume": 80, "Direction": "West"},
 {"Accident": "Accident 32", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 33": [
 {"Registration": "MH-765", "Owner": "Christopher Brown", "Type": "Bike"},
 {"Flow": "North", "Volume": 60, "Direction": "North"},
 {"Accident": "Accident 33", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 34": [
 {"Registration": "MH-098", "Owner": "Hannah White", "Type": "Car"},
 {"Flow": "South", "Volume": 90, "Direction": "South"},
 {"Accident": "Accident 34", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 35": [
 {"Registration": "MH-321", "Owner": "Isaac Black", "Type": "Truck"},
 {"Flow": "East", "Volume": 110, "Direction": "East"},
 {"Accident": "Accident 35", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 36": [
 {"Registration": "MH-654", "Owner": "Lillian Green", "Type": "Bike"},
 {"Flow": "West", "Volume": 35, "Direction": "West"},
 {"Accident": "Accident 36", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 37": [
 {"Registration": "MH-987", "Owner": "Nathan Brown", "Type": "Car"},
 {"Flow": "North", "Volume": 105, "Direction": "North"},
 {"Accident": "Accident 37", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 38": [
 {"Registration": "MH-210", "Owner": "Olivia White", "Type": "Truck"},
 {"Flow": "South", "Volume": 70, "Direction": "South"},
 {"Accident": "Accident 38", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 39": [
 {"Registration": "MH-543", "Owner": "Omar Black", "Type": "Bike"},
 {"Flow": "East", "Volume": 50, "Direction": "East"},
 {"Accident": "Accident 39", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 40": [
 {"Registration": "MH-876", "Owner": "Pamela Green", "Type": "Car"},
 {"Flow": "West", "Volume": 135, "Direction": "West"},
 {"Accident": "Accident 40", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 41": [
 {"Registration": "MH-109", "Owner": "Quinn Brown", "Type": "Truck"},
 {"Flow": "North", "Volume": 85, "Direction": "North"},
 {"Accident": "Accident 41", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 42": [
 {"Registration": "MH-432", "Owner": "Rachel White", "Type": "Bike"},
 {"Flow": "South", "Volume": 65, "Direction": "South"},
 {"Accident": "Accident 42", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 43": [
 {"Registration": "MH-765", "Owner": "Samuel Black", "Type": "Car"},
 {"Flow": "East", "Volume": 115, "Direction": "East"},
 {"Accident": "Accident 43", "Severity": "Minor", "Location": "Main Road"}
],
 "Vehicle 44": [
 {"Registration": "MH-098", "Owner": "Tina Green", "Type": "Truck"},
 {"Flow": "West", "Volume": 40, "Direction": "West"},
 {"Accident": "Accident 44", "Severity": "Major", "Location": "Main Road"}
],
 "Vehicle 45": [
 {"Registration": "MH-321", "Owner": "Tyler Brown", "Type": "Bike"},
 {"Flow": "North", "Volume": 100, "Direction": "North"},
 {"
```

## Task Description #8: Smart E-Commerce Platform – Data Structure

## Challenge

An e-commerce company wants to build a Smart Online Shopping System with:

1. Shopping Cart Management – Add and remove products dynamically.
2. Order Processing System – Orders processed in the order they are placed.
3. Top-Selling Products Tracker – Products ranked by sales count.
4. Product Search Engine – Fast lookup of products using product ID.
5. Delivery Route Planning – Connect warehouses and delivery locations.

## Student Task

- For each feature, select the most appropriate data structure from the list below:

- o Stack
- o Queue
- o Priority Queue
- o Linked List
- o Binary Search Tree (BST)
- o Graph
- o Hash Table
- o Deque

- Justify your choice in 2–3 sentences per feature.

- Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

- A table mapping feature → chosen data structure → justification.
- A functional Python program implementing the chosen feature with comments and docstrings.

PROMPT USED : ( FOR TABLE )

Generate a formatted table for a Smart E-Commerce Platform.

Features:

1. Shopping Cart Management
2. Order Processing System
3. Top-Selling Products Tracker
4. Product Search Engine
5. Delivery Route Planning

Choose the most appropriate data structure from:

Stack, Queue, Priority Queue, Linked List, Binary Search Tree (BST), Graph, Hash Table, Deque.

Requirements:

- Format as: Feature | Data Structure | Justification
- Provide 2–3 sentence justification per feature.
- Keep explanations short and clear.

PROMPT USED : (FOR CODE)

